Package 'manMetaVAR'

November 7, 2026

Title Meta-Analytic SEM for Multilevel Dynamics

Version 0.5.8

Description Research compendium for the manuscript
Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (Under Review).
From Person-Specific Dynamics to Population Inference:
A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework
for Integrating Complex Multilevel Dynamic Models.
<doi:10.0000/0000000000>.

URL https://github.com/jeksterslab/manMetaVAR,
https://jeksterslab.github.io/manMetaVAR/,
https://osf.io/uenr7/, https://doi.org/10.0000/0000000000

BugReports https://github.com/jeksterslab/manMetaVAR/issues

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Encoding UTF-8

LazyData true

LazyDataCompression xz

Roxygen list(markdown = TRUE)

Depends R (>= 4.5.0), OpenMx (>= 2.22.11), fitVARMxID (>= 1.0.6),
metaDyn (>= 1.0.4)

LinkingTo Rcpp, RcppArmadillo

Imports posterior, Rcpp, stats, utils, simStateSpace (>= 1.2.16),
ggplot2

Remotes jeksterslab/fitVARMxID, jeksterslab/metaDyn

Suggests knitr, rmarkdown, testthat

Config/roxygen2/version 8.1.1.9000

NeedsCompilation yes

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`calibration` *Calibration of Stage 1 Sampling Variances in the Simulation*

Description

For each simulation condition and transition parameter, the between-person variance of the Stage 1 estimates, the mean Stage 1 sampling variance, and their ratio, averaged over admissible MetaVAR replications (`SumStage1Calibration()`). Under zero transition heterogeneity, a ratio of one means that the Stage 1 sampling variances are calibrated; a ratio above one means that they are too small.

Usage

```
data(calibration)
```

Format

A data frame with one row per task ID and transition parameter:

taskid Simulation task ID.

replications Number of replications.

replications_used Number of admissible MetaVAR replications.

n Number of persons.

time Number of measurement occasions; `NA` denotes the unbalanced condition.

heterogeneity Transition-parameter heterogeneity multiplier.

position Position of the transition parameter in the Stage 1 parameter vector (3 to 6 for a_{11}, a_{21}, a_{12}, a_{22}).

between_var Between-person variance of the Stage 1 estimates.

mean_sampling_var Mean Stage 1 sampling variance.

ratio `between_var` divided by `mean_sampling_var`.

Author(s)

Ivan Jacob Agaloos Pesigan

`Check` *Check Replication*

Description

Check Replication

Usage

```
Check(taskid, repid, output_folder, naive, metavar, mplus)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`naive`	Logical. If `naive = TRUE`, fit the naive approach.
`metavar`	Logical. If `metavar = TRUE`, fit the model using the `metaDyn` package.
`mplus`	Logical. If `mplus = TRUE`, fit the model using DSEM in `Mplus`.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`CheckFitDTVAR` *Check Replication - FitDTVAR*

Description

Check Replication - FitDTVAR

Usage

```
CheckFitDTVAR(taskid, repid, output_folder, suffix)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
suffix	Character string. Simulation file suffix returned by .SimSuffix().

Details

This function is executed via the Check function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

CheckFitMetaVAR *Check Replication - FitMetaVAR*

Description

Check Replication - FitMetaVAR

Usage

```
CheckFitMetaVAR(taskid, repid, output_folder, suffix)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
suffix	Character string. Simulation file suffix returned by .SimSuffix().

Details

This function is executed via the Check function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

CheckFitMplus *Check Replication - FitMplus*

Description

Check Replication - FitMplus

Usage

```
CheckFitMplus(taskid, repid, output_folder, suffix)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
suffix	Character string. Simulation file suffix returned by `.SimSuffix()`.

Details

This function is executed via the `Check` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

CheckFitMplusPriors *Check Replication - FitMplus (User Defined Priors)*

Description

Check Replication - FitMplus (User Defined Priors)

Usage

```
CheckFitMplusPriors(taskid, repid, output_folder, suffix)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
suffix	Character string. Simulation file suffix returned by .SimSuffix().

Details

This function is executed via the Check function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

CheckFitNaive *Check Replication - FitNaive*

Description

Check Replication - FitNaive

Usage

```
CheckFitNaive(taskid, repid, output_folder, suffix)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
suffix	Character string. Simulation file suffix returned by .SimSuffix().

Details

This function is executed via the Check function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

Compress *Compress Replication*

Description

Compress Replication

Usage

```
Compress(taskid, repid, output_folder)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

diagnostics *Bayesian Mplus Diagnostics Across Simulation Conditions*

Description

Parameter- and run-level Bayesian Mplus diagnostics for all simulation task IDs, stored separately for the default and alternative prior specifications.

Usage

```
data(diagnostics_default)
data(diagnostics_priors)
```

Format

A list with three elements:

parameters A data frame with one row per task ID, replication, prior condition, and Mplus parameter. It contains the simulation design variables, posterior SD, R-hat, bulk and tail ESS, and MCSEs.

runs A data frame with one row per task ID, replication, and prior condition containing run-level diagnostics and elapsed time.

replications Number of simulation replications represented in each task/prior cell.

The object is generated by `.setup/data-process/data-process-005-diagnostics.R` from the task-specific outputs of `SumFitMplusDiagnostics()` and `SumFitMplusPriorsDiagnostics()`.

Details

Each bundle covers 36 bivariate tasks and one prior branch, with 1,000 replications per task/prior cell: 36,000 run rows and 792,000 parameter rows (22 parameters per run). Use `LoadMplusDiagnostics()` to combine the bundles into 72,000 run rows and 1,584,000 parameter rows. The separate bundles replace `data(diagnostics)` and the earlier 10-replication bundle. Block-level failures differ from full-run gates. Primary thresholds are Rhat >1.01, bulk/tail ESS <400, MCSE/SD >0.05; nonfinite diagnostics fail. These are archived summaries, not new fits.

The compact parameter tables omit posterior means, medians, and interval endpoints (`mean`, `median`, `ll`, and `ul`) to reduce bundle size. Posterior SD (`sd`), R-hat, bulk and tail ESS, and all four MCSE columns are retained without rounding or subsampling. Run diagnostics and replication counts are unchanged. The complete posterior summaries remain in the task-specific source files.

Author(s)

Ivan Jacob Agaloos Pesigan

---

diagnostics_criteria *DSEM Posterior-Quality Failure Rates Under Three Criteria*

Description

Proportions of DSEM replications that fail the posterior-quality criteria, by parameter group and prior, computed with `TabMplusDiagSensitivity()` under strict ($\hat{R} \leq 1.01$, ESS ≥ 400, MCSE/SD $\leq .05$), moderate ($\hat{R} \leq 1.05$, ESS ≥ 100, MCSE/SD $\leq .10$), and lenient ($\hat{R} \leq 1.10$, ESS ≥ 100) criteria.

Usage

```
data(diagnostics_criteria)
```

Format

A data frame with one row per design condition and criterion:

taskid Simulation task ID.

case_label Design-condition label.

n Number of persons.

time Number of measurement occasions; `NA` denotes the unbalanced condition.

heterogeneity Transition-parameter heterogeneity multiplier.

Innovation covariance | Default, Innovation covariance | User, Fixed effects | Default, Fixed effects | User, Random ef Proportion of replications failing the criterion, by parameter group and prior.

criterion `"strict"`, `"moderate"`, or `"lenient"`.

rhat_threshold, ess_threshold, mcse_threshold Thresholds of the criterion.

Author(s)

Ivan Jacob Agaloos Pesigan

`dsem_converged` *DSEM Recovery Over All and Over Converged Replications*

Description

Bias, root mean square error, and coverage of the DSEM estimates in each simulation condition, computed over all replications and over the replications in which every parameter meets the moderate posterior-quality criterion (finite diagnostics, $\hat{R} \leq 1.05$, bulk and tail ESS ≥ 100, each MCSE/SD $\leq .10$). The kept replications are a selected subset.

Usage

```
data(dsem_converged)
```

Format

A data frame with one row per task ID, prior, and parameter:

taskid Simulation task ID.

method `"BMLVAR-Default"` or `"BMLVAR-Priors"`.

n Number of persons.

time Number of measurement occasions; `NA` denotes the unbalanced condition.

heterogeneity Transition-parameter heterogeneity multiplier.

replications Number of replications.

replications_kept Number of replications meeting the moderate criterion.

parnames Parameter name.

parameter Calibrated population value.

bias_all, rmse_all, coverage_all Bias, root mean square error, and coverage of the 95 percent credible interval over all replications.

bias_kept, rmse_kept, coverage_kept The same over the kept replications; `NA` when none is kept.

Author(s)

Ivan Jacob Agaloos Pesigan

`FigMetricHeatmap` *Results Heatmap*

Description

Plot results heatmap.

Usage

```
FigMetricHeatmap(results, metric_name, grey_scale = FALSE, values = FALSE)
```

Arguments

`results`	`results` data frame.
`metric_name`	Character string. `"coverage"` for coverage probability, `"abs_bias"` for absolute bias on the parameter scale, `"abs_rel_bias"` for the legacy mixed relative/absolute bias display, `"rmse"` for RMSE, `"power"` for statistical power, and `"type1_error"` for the Type I error rate.
`grey_scale`	Logical. If `TRUE`, use a grey-scale fill suitable for black-and-white printing.
`values`	Logical. If `TRUE`, display values inside tiles.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: `FigMplusDiagnosticsHeatmap()`, `FigMplusDiagnosticsOverview()`, `FigOverview()`

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigMetricHeatmap(
  results = results,
  metric_name = "coverage"
)

## End(Not run)
```

`FigMplusDiagnosticsHeatmap` *Bayesian Diagnostic Heatmaps Across Simulation Conditions*

Description

Summarize parameter-level Mplus Bayesian diagnostics across all simulation task IDs and display them using the same design-cell-by-parameter layout as `FigMetricHeatmap()`. The function can display the two prior conditions directly or paired user-prior minus default-prior differences.

Usage

```
FigMplusDiagnosticsHeatmap(
  diagnostics,
  metric_name = "failure_rate",
  comparison = "difference",
  diagnostic = NULL,
  taskid = NULL,
  parm = NULL,
  center = "mean",
  rhat_threshold = 1.01,
  ess_threshold = 400,
  mcse_threshold = 0.05,
  relative_mcse = TRUE,
  grey_scale = FALSE,
  values = FALSE
)
```

Arguments

`diagnostics`	An all-task diagnostic object containing a `parameters` data frame, such as the `diagnostics` object generated by `.setup/data-process/data-process-z-diagnostics.R`. A parameter-level data frame may also be supplied directly.
`metric_name`	Character string. Valid values are `"failure_rate"`, `"rhat"`, `"ess"`, and `"mcse"`.
`comparison`	Character string. Use `"levels"` to show default and user-specified priors or `"difference"` to show paired user-prior minus default-prior differences.

diagnostic	Optional diagnostic panel names to retain. If NULL, all panels for the selected metric are displayed.
taskid	Optional positive integer task IDs to include. The default is to summarize all task IDs.
parm	Parameter names or positive integer positions to include. If NULL, all k = 2 Mplus parameters are included.
center	Character string indicating whether continuous diagnostics are summarized using the "mean" or "median" across replications. Failure rates always use the mean.
rhat_threshold	Numeric R-hat threshold.
ess_threshold	Numeric bulk and tail ESS threshold.
mcse_threshold	Numeric MCSE threshold.
relative_mcse	Logical. If TRUE, Monte Carlo standard errors are divided by the posterior standard deviation before summarizing.
grey_scale	Logical. If TRUE, use a grey-scale fill.
values	Logical. If TRUE, print summarized values inside cells.

Details

Failure rates count nonfinite diagnostic values as failures. For comparison = "difference", diagnostic values are paired by task ID, replication, parameter, and diagnostic before computing user-prior minus default-prior differences. Blue cells indicate improvement under user priors and red cells indicate deterioration.

Value

A ggplot object. Its data element contains the all-task summary used in the heatmap.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: FigMetricHeatmap(), FigMplusDiagnosticsOverview(), FigOverview()

Examples

```
## Not run:
data(diagnostics, package = "manMetaVAR")

FigMplusDiagnosticsHeatmap(
  diagnostics = diagnostics,
  metric_name = "failure_rate",
  comparison = "difference"
)

FigMplusDiagnosticsHeatmap(
```

```
  diagnostics = diagnostics,
  metric_name = "rhat",
  comparison = "levels",
  values = TRUE
)

## End(Not run)
```

`FigMplusDiagnosticsOverview` *Compact Bayesian Diagnostic Overview Across Simulation Conditions*

Description

Summarize Mplus Bayesian diagnostic failures across all simulation task IDs at the Innovation, fixed-effect, and random-effect block level. This compact overview is intended for manuscript figures and reviewer-oriented prior sensitivity analyses. Use `FigMplusDiagnosticsHeatmap()` for parameter-level drill-down figures.

Usage

```
FigMplusDiagnosticsOverview(
  diagnostics,
  comparison = "difference",
  diagnostic = "Any diagnostic",
  unit = "replication",
  taskid = NULL,
  target = NULL,
  rhat_threshold = 1.01,
  ess_threshold = 400,
  mcse_threshold = 0.05,
  relative_mcse = TRUE,
  grey_scale = FALSE,
  values = FALSE,
  label_threshold = 0.05
)
```

Arguments

`diagnostics`	An all-task diagnostics object containing a `parameters` data frame, or the parameter-level data frame itself.
`comparison`	Character string. Use `"levels"` to show diagnostic failure rates under each prior specification or `"difference"` to show paired user-prior minus default-prior differences.

diagnostic	Diagnostic failure panels to display. Available values are "Any diagnostic", "R-hat", "Bulk ESS", "Tail ESS", and "Relative MCSE" when relative_mcse = TRUE.
unit	Character string controlling the unit summarized within each design cell. "replication" classifies a replication as failed when any parameter in the selected parameter block fails. "parameter" averages failure indicators across parameter-replication pairs.
taskid	Optional positive integer task IDs to include.
target	Optional parameter blocks to include. Available values are "Innovation", "FE", and "RE".
rhat_threshold	Numeric R-hat threshold.
ess_threshold	Numeric bulk and tail ESS threshold.
mcse_threshold	Numeric MCSE threshold.
relative_mcse	Logical. If TRUE, Monte Carlo standard errors are divided by the posterior standard deviation before classifying failures.
grey_scale	Logical. If TRUE, use grey-scale fills.
values	Logical. If TRUE, print values inside sufficiently large cells.
label_threshold	Numeric value between zero and one. Labels with an absolute value smaller than this threshold are omitted. For level plots, the threshold applies directly to the failure rate.

Details

The default unit = "replication" answers a reviewer-friendly question: what proportion of simulation replications had at least one diagnostic problem within each parameter block? For comparison = "difference", the function pairs prior specifications within task ID and replication before calculating user-prior minus default-prior differences. Negative values indicate fewer failures under user-specified priors.

Value

A ggplot object. Its data element contains the block-level summaries used in the figure.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: FigMetricHeatmap(), FigMplusDiagnosticsHeatmap(), FigOverview()

Examples

```
## Not run:
data(diagnostics, package = "manMetaVAR")

FigMplusDiagnosticsOverview(
  diagnostics = diagnostics,
  comparison = "levels"
)

FigMplusDiagnosticsOverview(
  diagnostics = diagnostics,
  comparison = "difference",
  values = TRUE
)

## End(Not run)
```

`FigOverview` *Results Overview*

Description

Plot results overview.

Usage

```
FigOverview(results)
```

Arguments

`results`	`results` data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: `FigMetricHeatmap()`, `FigMplusDiagnosticsHeatmap()`, `FigMplusDiagnosticsOverview()`

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigOverview(results)

## End(Not run)
```

FitDTVAR *Fit the Model using the fitVARMxID Package*

Description

The function fits the model using the fitVARMxID package.

Usage

```
FitDTVAR(data, ncores = NULL, seed = NULL)
```

Arguments

data	R object. Output of the GenData() function.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: FitMetaVAR(), FitMplus(), FitMplusDiagnostics(), FitNaive(), LRTMetaVAR(), MplusInput()

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

FitMetaVAR *Multivariate Meta-Analysis using the metaDyn Package*

Description

The function performs multivariate meta-snalysis using the metaDyn package.

Usage

```
FitMetaVAR(fit, ncores = NULL, seed = NULL)
```

Arguments

fit	R object. Output of the FitDTVAR() function.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: FitDTVAR(), FitMplus(), FitMplusDiagnostics(), FitNaive(), LRTMetaVAR(), MplusInput()

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
meta <- FitMetaVAR(fit = fit, seed = seed)
summary(meta)
print(meta)
coef(meta)
vcov(meta)

## End(Not run)
```

FitMplus *Fit the Model using Mplus*

Description

The function fits the model using Mplus.

Usage

```
FitMplus(
  data,
  chains = 2L,
  iter = 40000L,
  fscores = NULL,
  plot = FALSE,
  default_priors = TRUE,
  wd = ".",
  mplus_bin = NULL,
  ncores = NULL,
  seed = NULL
)
```

Arguments

data	R object. Output of the GenData() function.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.
default_priors	Logical. If default_priors = TRUE, use Mplus default priors.
wd	Character string. Working directory.
mplus_bin	Character string. Path to Mplus binary. If mplus_bin = NULL, the function will try to find the appropriate binary.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: FitDTVAR(), FitMetaVAR(), FitMplusDiagnostics(), FitNaive(), LRTMetaVAR(), MplusInput()

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
dsem <- FitMplus(data = data)
print(dsem)
summary(dsem)
coef(dsem)
vcov(dsem)
confint(dsem)
plot(dsem)
plot(dsem, what = "trace")

## End(Not run)
```

FitMplusDiagnostics *Posterior Diagnostics for FitMplus*

Description

The function computes parameter-level posterior summaries and Markov chain Monte Carlo diagnostics from the posterior draws saved by FitMplus().

Usage

```
FitMplusDiagnostics(object, burnin = NULL, level = 0.95)
```

Arguments

object	Object of class manmetavar.mplus returned by FitMplus().
burnin	Integer indicating the number of initial draws to discard from each chain. If burnin = NULL, use object$burnin.
level	Numeric value indicating the credibility level.

Value

An object of class manmetavar.mplus.diagnostics containing a parameter-level diagnostics data frame and a run-level diagnostics data frame.

See Also

Other Model Fitting Functions: FitDTVAR(), FitMetaVAR(), FitMplus(), FitNaive(), LRTMetaVAR(), MplusInput()

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitMplus(data = data, seed = seed)
diagnostics <- FitMplusDiagnostics(fit)
diagnostics$parameters
diagnostics$run

## End(Not run)
```

FitNaive *Naive*

Description

The function performs the naive "fit-many-then-summarize".

Usage

```
FitNaive(fit, seed = NULL)
```

Arguments

fit	R object. Output of the FitDTVAR() function.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: FitDTVAR(), FitMetaVAR(), FitMplus(), FitMplusDiagnostics(), LRTMetaVAR(), MplusInput()

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
naive <- FitNaive(fit = fit, seed = seed)
summary(naive)
print(naive)
coef(naive)
vcov(naive)

## End(Not run)
```

`GenData` *Simulate Data*

Description

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

Usage

```
GenData(taskid, seed = NULL)
```

Arguments

`taskid`	Positive integer. Task ID.
`seed`	Integer. Random seed.

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
print(data)
summary(data)
plot(data)

## End(Not run)
```

LoadMplusDiagnostics *Load Both Prior Conditions of the Bundled Mplus Diagnostics*

Description

Loads and combines `diagnostics_default` and `diagnostics_priors` for diagnostic plots and comparisons between prior specifications.

Usage

```
LoadMplusDiagnostics()
```

Value

A `manmetavar.mplus.diagnostics.all` object containing parameter and run diagnostics for both prior conditions, in the original combined task, method, replication, and parameter order.

Examples

```
## Not run:
diagnostics <- LoadMplusDiagnostics()
FigMplusDiagnosticsHeatmap(diagnostics, comparison = "difference")

## End(Not run)
```

lrt *Boundary Likelihood Ratio Test in the Simulation*

Description

Rejection rates of the MetaVAR boundary likelihood ratio test (`LRTMetaVAR()`) of zero transition heterogeneity in each simulation condition, at $\alpha = .05$, with the simulated chi-bar-square and the naive chi-square references. At zero heterogeneity, the rejection rates are Type I error rates; otherwise, they are power.

Usage

```
data(lrt)
```

Format

A data frame with one row per task ID:

taskid Simulation task ID.

replications Number of replications.

replications_used Number of admissible MetaVAR replications.

method "MetaVAR".

n Number of persons.

time Number of measurement occasions; NA denotes the unbalanced condition.

heterogeneity Transition-parameter heterogeneity multiplier.

df Number of tested parameters.

negative_stat Number of replications with a statistic below -10^{-6}.

null_status_nonzero Number of replications whose null fit had a nonzero status code.

reject_chibar Rejection rate with the chi-bar-square reference.

reject_chisq Rejection rate with the chi-square reference.

crit_chibar Median simulated 95th percentile of the chi-bar-square reference.

Author(s)

Ivan Jacob Agaloos Pesigan

LRTMetaVAR *Boundary Likelihood Ratio Test for Transition Heterogeneity*

Description

Tests whether the between-person covariance matrix of the transition parameters is zero in a fitted MetaVAR Stage 2 model. The model is refitted with the transition rows and columns of $\boldsymbol{\tau}^2$ fixed at zero (variances at the lower bound of the softplus parameterization), and the likelihood ratio statistic is referred to a simulated chi-bar-square distribution for the cone of positive semidefinite matrices, using the Fisher information at the null fit, and to the naive chi-square distribution.

Usage

```
LRTMetaVAR(
  fit,
  block = 3:6,
  others = 1:2,
  cross = FALSE,
  draws = 2000L,
  seed = NULL
)
```

Arguments

fit	Fitted MetaVAR Stage 2 model: an object returned by FitMetaVAR(), by metaDyn::MetaVARMx(), or the underlying MxModel.
block	Integer vector. Positions of the transition parameters in $\boldsymbol{\tau}^2$.
others	Integer vector. Positions of the remaining random effects in $\boldsymbol{\tau}^2$.

cross	Logical. If cross = TRUE, the covariances between block and others are free in fit and are also set to zero under the null hypothesis.
draws	Positive integer. Number of draws from the chi-bar-square distribution.
seed	Integer. Random seed for the chi-bar-square draws.

Value

A one-row data frame with the -2 log-likelihoods of the full and null models, the statistic, the null-fit status code, the number of tested parameters (df), the chi-bar-square p-value and 95% critical value, and the naive chi-square p-value.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Model Fitting Functions: FitDTVAR(), FitMetaVAR(), FitMplus(), FitMplusDiagnostics(), FitNaive(), MplusInput()

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 25, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
meta <- FitMetaVAR(fit = fit, seed = seed)
LRTMetaVAR(fit = meta, seed = seed)

## End(Not run)
```

---

manmetavar-data-methods

Methods for Objects of Class manmetavar.data

Description

This page documents the available methods for objects of class manmetavar.data.

Usage

```
## S3 method for class 'manmetavar.data'
print(x, ...)

## S3 method for class 'manmetavar.data'
summary(object, ...)

## S3 method for class 'manmetavar.data'
plot(x, ...)
```

Arguments

x	Object of class manmetavar.data.
...	additional arguments.
object	Object of class manmetavar.data.

Author(s)

Ivan Jacob Agaloos Pesigan

---

manmetavar-dtvar-methods

Methods for Objects of Class manmetavar.dtvar

Description

This page documents the available methods for objects of class manmetavar.dtvar.

Usage

```
## S3 method for class 'manmetavar.dtvar'
coef(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  ncores = NULL,
  ...
)

## S3 method for class 'manmetavar.dtvar'
vcov(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  robust = FALSE,
  ncores = NULL,
  ...
)
```

```
## S3 method for class 'manmetavar.dtvar'
print(
  x,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  digits = 4,
  ncores = NULL,
  ...
)

## S3 method for class 'manmetavar.dtvar'
summary(
  object,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  digits = 4,
  ncores = NULL,
  ...
)
```

Arguments

object	Object of class manmetavar.dtvar.
mu	Logical. If mu = TRUE, include estimates of the mu vector, if available. If mu = FALSE, exclude estimates of the mu vector.
alpha	Logical. If alpha = TRUE, include estimates of the alpha vector, if available. If alpha = FALSE, exclude estimates of the alpha vector.
beta	Logical. If beta = TRUE, include estimates of the beta matrix, if available. If beta = FALSE, exclude estimates of the beta matrix.
nu	Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.
psi	Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.
theta	Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix.
ncores	Positive integer. Number of cores to use.

...	additional arguments.
robust	Logical. If TRUE, use robust (sandwich) sampling variance-covariance matrix. If FALSE, use normal theory sampling variance-covariance matrix.
x	Object of class manmetavar.dtvar.
means	Logical. If means = TRUE, return means. Otherwise, the function returns raw estimates.
digits	Integer indicating the number of decimal places to display.

Author(s)

Ivan Jacob Agaloos Pesigan

---

manmetavar-metavar-methods

Methods for Objects of Class manmetavar.metavar

Description

This page documents the available methods for objects of class manmetavar.metavar.

This page documents the available methods for objects of class manmetavar.naive.

Usage

```
## S3 method for class 'manmetavar.metavar'
coef(object, ...)

## S3 method for class 'manmetavar.metavar'
vcov(object, robust = NULL, ...)

## S3 method for class 'manmetavar.metavar'
print(x, alpha = NULL, robust = NULL, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
summary(object, alpha = NULL, robust = NULL, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
confint(object, parm = NULL, level = 0.95, robust = NULL, ...)

## S3 method for class 'manmetavar.naive'
coef(object, ...)

## S3 method for class 'manmetavar.naive'
vcov(object, ...)

## S3 method for class 'manmetavar.naive'
```

```
print(x, ...)

## S3 method for class 'manmetavar.naive'
summary(object, alpha = 0.05, digits = 4, ...)
```

Arguments

object	Object of class manmetavar.naive.
...	additional arguments.
robust	Logical. If TRUE, use robust (sandwich) sampling variance-covariance matrix. If FALSE, use normal theory sampling variance-covariance matrix. If NULL, the function will check object if robust standard errors are available.
x	Object of class manmetavar.naive.
alpha	Numeric vector. Significance level α.
digits	Integer indicating the number of decimal places to display.
parm	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
level	the confidence level required.

Author(s)

Ivan Jacob Agaloos Pesigan

---

manmetavar-mplus-methods

Methods for Objects of Class manmetavar.mplus

Description

This page documents the available methods for objects of class manmetavar.mplus.

Usage

```
## S3 method for class 'manmetavar.mplus'
coef(object, median = TRUE, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
vcov(object, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
summary(object, alpha = 0.05, median = TRUE, digits = 4, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
confint(object, parm = NULL, level = 0.95, burnin = NULL, ...)
```

```
## S3 method for class 'manmetavar.mplus'
plot(
  x,
  what = "posterior",
  parm = NULL,
  level = 0.95,
  burnin = NULL,
  legend_loc = "topright",
  ...
)
```

Arguments

object	Object of class manmetavar.mplus.
median	Logical. If median = TRUE, return median of the posterior. If median = FALSE, return mean of the posterior.
burnin	Integer indicating initial samples to discard. If burnin = NULL, use the burn-in stored in object$burnin.
...	additional arguments.
alpha	Numeric vector. Significance level α.
digits	Integer indicating the number of decimal places to display.
parm	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
level	the confidence level required.
x	Object of class manmetavar.mplus.
what	Character string. If what = "posterior", return posterior distribution plots. If what = "trace", return trace plots. If what = "rhat", "ess", or "mcse", return the corresponding Bayesian diagnostic plot from FitMplusDiagnostics().
legend_loc	Character string. Legend location.

Author(s)

Ivan Jacob Agaloos Pesigan

---

model *Model Parameters*

Description

Model Parameters

Usage

```
data(model)
```

Format

A list with 15 elements:

k Number of variables.

mu_mu Mean of the set-point vector $\boldsymbol{\mu}$.

mu_sigma Covariance matrix of the parameter $\boldsymbol{\mu}$.

mu_sigma_l Cholesky factor of the covariance matrix of the parameter $\boldsymbol{\mu}$.

beta_mu Mean of lagged coefficients matrix $\boldsymbol{\beta}$.

beta_sigma Covariance matrix of the parameter $\text{vec}\,(\boldsymbol{\beta})$.

beta_sigma_l Cholesky factor of the covariance matrix of the parameter $\text{vec}\,(\boldsymbol{\beta})$.

psi Process noise covariance matrix $\boldsymbol{\Psi}$.

psi_l Cholesky factor of the process noise covariance matrix $\boldsymbol{\Psi}$.

psi_d_ldl `uc_d` of the LDL' decomposition of the process noise covariance matrix $\boldsymbol{\Psi}$. See `fitVARMxID::LDL()`.

psi_l_ldl `s_l` of the LDL' decomposition of the process noise covariance matrix $\boldsymbol{\Psi}$. See `fitVARMxID::LDL()`.

ma_fixed Vector of fixed effects $\boldsymbol{\theta} = [\boldsymbol{\mu}, \text{vec}\,(\boldsymbol{\beta})]'$

ma_random Matrix of random effects $\boldsymbol{\theta} = [\boldsymbol{\mu}, \text{vec}\,(\boldsymbol{\beta})]'$

ma_random_d_ldl `uc_d` of the LDL' decomposition of the random-effects covariance. See `fitVARMxID::LDL()`.

ma_random_l_ldl `s_l` of the LDL' decomposition of the random-effects covariance. See `fitVARMxID::LDL()`.

Author(s)

Ivan Jacob Agaloos Pesigan

MplusInput *Create Mplus Input File*

Description

The function creates an Mplus input file.

Usage

```
MplusInput(
  fn_data,
  fn_posterior,
  fn_factorscores,
  chains,
  iter,
  fscores,
  plot,
  default_priors = TRUE,
  ncores = NULL,
  seed = NULL
)
```

Arguments

fn_data	Character string. Filename for data file.
fn_posterior	Character string. Filename for posterior output.
fn_factorscores	Character string. Filename for factor scores output.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.
default_priors	Logical. If default_priors = TRUE, use Mplus default priors.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: FitDTVAR(), FitMetaVAR(), FitMplus(), FitMplusDiagnostics(), FitNaive(), LRTMetaVAR()

Examples

```
cat(
  MplusInput(
    fn_data = "data.dat",
    fn_posterior = "posterior.dat",
    fn_factorscores = "factorscores.dat",
    chains = 2L,
    iter = 40000L,
    fscores = 1000L,
    plot = TRUE,
    default_priors = TRUE,
    ncores = NULL,
    seed = 42
```

```
  )
)
```

`params` *Simulation Parameters*

Description

Simulation Parameters

Usage

```
data(params)
```

Format

A dataframe with 12 rows and 3 columns:

taskid Simulation Task ID.

n Sample size.

time Number of measurement occassions. If `NA`, measurement occasions vary across IDs.

heterogeneity Transition-parameter heterogeneity multiplier: 0, 1, or 2.

Author(s)

Ivan Jacob Agaloos Pesigan

`population` *Calibrated Population Parameters*

Description

Population moments implied by the stationary transition-matrix generator. The moments of the set points are analytical. The moments of the transition matrices are calculated by Monte Carlo integration over the stationary region for each nonzero heterogeneity condition.

Usage

```
data(population)
```

Format

A list with two elements:

conditions Named list indexed by heterogeneity. Each element contains the calibrated means and covariance matrices, a named population parameter vector, and Monte Carlo calibration information.

calibration Metadata describing the generator, package version, random seed, population size, batch size, stationarity margin, and generation time.

Author(s)

Ivan Jacob Agaloos Pesigan

`results` *Simulation Results*

Description

Simulation Results

Usage

```
data(results)
```

Format

A data frame with one row per design condition, method, interval type, and parameter:

taskid Simulation task ID.

replications Number of replications.

replications_used Number of admissible replications contributing to the summary.

success_rate Proportion of requested replications that were admissible and contributed to the summary.

parnames Parameter name.

parameter Calibrated population parameter value. For transition parameters under stability rejection, this is the realized truncated population value rather than the nominal untruncated value.

method Estimation method: MetaVAR, BMLVAR with default priors, BMLVAR with user-defined priors, or the uncertainty-uncorrected method.

n Number of persons.

time Number of measurement occasions; `NA` denotes the unbalanced condition.

heterogeneity Transition-parameter heterogeneity multiplier: 0, 1, or 2.

ci Confidence or credible interval method.

est Mean parameter estimate.

se Mean standard error or posterior standard deviation.

z Mean test statistic when available.

p Mean p-value when available.

ll Mean lower interval limit.

ul Mean upper interval limit.

sig Proportion statistically significant when available.

zero_hit Proportion of intervals containing zero.

theta_hit Proportion of intervals containing the calibrated population value.

sq_error Mean squared error.

bias Mean estimate minus the calibrated population value.

rel_bias Relative bias for nonzero population values and `NA` when the population value is zero.

rmse Root mean squared error.

n_bias Number of finite replication-level bias values used to calculate the Monte Carlo standard error of bias.

n_rmse Number of finite replication-level squared errors used to calculate the Monte Carlo standard error of RMSE.

n_coverage Number of finite coverage indicators used to calculate the Monte Carlo standard error of coverage.

n_rejection Number of finite rejection indicators used to calculate the Monte Carlo standard error of the rejection rate.

mcse_bias Monte Carlo standard error of bias.

mcse_rmse Delta-method Monte Carlo standard error of RMSE, obtained from the Monte Carlo standard error of mean squared error.

coverage Coverage probability.

mcse_coverage Binomial Monte Carlo standard error of coverage.

rejection_rate Proportion of intervals excluding zero.

mcse_rejection_rate Binomial Monte Carlo standard error of the rejection rate.

power Rejection rate for nonzero population values and `NA` for zero population values.

mcse_power Monte Carlo standard error of power for nonzero population values and `NA` for zero population values.

type1_error Rejection rate for zero population values and `NA` for nonzero population values.

mcse_type1_error Monte Carlo standard error of the Type I error rate for zero population values and `NA` for nonzero population values.

par_idx Parameter display index.

Author(s)

Ivan Jacob Agaloos Pesigan

Sim *Simulation Replication*

Description

Simulation Replication

Usage

```
Sim(
  taskid,
  repid,
  output_folder,
  overwrite,
  integrity,
  seed,
  data,
  naive,
  metavar,
  mplus,
  chains,
  iter,
  fscores,
  plot
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
seed	Integer. Random seed.
data	Logical. Simulate data.
naive	Logical. If naive = TRUE, fit the naive approach.
metavar	Logical. If metavar = TRUE, fit the model using the metaDyn package.
mplus	Logical. If mplus = TRUE, fit the model using DSEM in Mplus.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFitDTVAR` *Simulation Replication - FitDTVAR*

Description

Simulation Replication - FitDTVAR

Usage

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMetaVAR *Simulation Replication - FitMetaVAR*

Description

Simulation Replication - FitMetaVAR

Usage

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Simulation file suffix returned by .SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplus *Simulation Replication - FitMplus*

Description

Simulation Replication - FitMplus

Usage

```
SimFitMplus(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  chains,
  iter,
  fscores,
  plot
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Simulation file suffix returned by .SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFitMplusDiagnostics`
Simulation Replication - FitMplus Diagnostics

Description

Simulation Replication - FitMplus Diagnostics

Usage

```
SimFitMplusDiagnostics(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplusPriors *Simulation Replication - FitMplus (User Defined Priors)*

Description

Simulation Replication - FitMplus (User Defined Priors)

Usage

```
SimFitMplusPriors(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  chains,
  iter,
  fscores,
  plot
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Simulation file suffix returned by .SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFitMplusPriorsDiagnostics`
Simulation Replication - FitMplus Diagnostics (User Defined Priors)

Description

Simulation Replication - FitMplus Diagnostics (User Defined Priors)

Usage

```
SimFitMplusPriorsDiagnostics(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitNaive *Simulation Replication - FitNaive*

Description

Simulation Replication - FitNaive

Usage

```
SimFitNaive(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Simulation file suffix returned by .SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFN *Simulation File Name*

Description

Simulation File Name

Usage

```
SimFN(output_type, output_folder, suffix)
```

Arguments

output_type	Character string. Output type. Valid values include "data", "fit-dt-var-mx", "fit-meta-var-mx", "fit-mplus", and "fit-naive".
output_folder	Character string. Output folder.
suffix	Character string. Simulation file suffix returned by .SimSuffix().

Value

Returns a character string file name with the output_folder in the OS-specific format.

SimGenData *Simulation Replication - GenData*

Description

Simulation Replication - GenData

Usage

```
SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Simulation file suffix returned by .SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimProj` *Simulation Project Name*

Description

Simulation Project Name

Usage

```
SimProj()
```

Value

Returns the project name as a character string.

Author(s)

Ivan Jacob Agaloos Pesigan

`simulation_diagnostics` *Simulation Diagnostics*

Description

Simulation diagnostics generated from the machine-readable replication status manifests. The objects contain convergence/admissibility counts, boundary or near-zero heterogeneity diagnostics, and runtime summaries.

Usage

```
data(simulation_diagnostics)
```

Format

A list with components `thresholds`, `status`, `status_summary`, `boundary_parameter_replications`, `boundary_parameter_summary`, `boundary_replication_diagnostics`, `boundary_replication_summary`, `runtime_replications`, and `runtime_summary`.

`Sum` *Summary*

Description

Summary

Usage

```
Sum(
  taskid,
  reps,
  output_folder,
  overwrite,
  integrity,
  naive,
  metavar_normal,
  metavar_robust,
  mplus,
  ncores,
  lrt = FALSE,
  calibration = FALSE
)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`naive`	Logical. If `naive = TRUE`, summarize naive estimates.
`metavar_normal`	Logical. If `metavar_normal = TRUE`, summarize metaVAR model using normal confidence intervals.
`metavar_robust`	Logical. If `metavar_robust = TRUE`, summarize metaVAR model using robust (sandwich) confidence intervals.
`mplus`	Logical. If `mplus = TRUE`, summarize the DSEM model.
`ncores`	Positive integer. Number of cores to use.

`lrt`	Logical. If `lrt = TRUE`, run the MetaVAR boundary likelihood ratio test (`SumFitMetaVARLRT()`).
`calibration`	Logical. If `calibration = TRUE`, summarize the calibration of the Stage 1 sampling variances (`SumStage1Calibration()`).

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SumDiagnostics` *Summarize Simulation Diagnostics*

Description

Summarizes the machine-readable status manifests, boundary or near-zero heterogeneity diagnostics, and runtime for a simulation task.

Usage

```
SumDiagnostics(
  taskid,
  reps,
  output_folder,
  overwrite,
  integrity,
  naive,
  metavar,
  mplus,
  variance_tol = 1e-06,
  eigen_tol = 1e-08
)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`naive`	Logical. If `naive = TRUE`, fit the naive approach.
`metavar`	Logical. If `metavar = TRUE`, fit the model using the `metaDyn` package.

mplus	Logical. If mplus = TRUE, fit the model using DSEM in Mplus.
variance_tol	Numeric scalar. Variance estimates no greater than this value are classified as boundary estimates for ML-based methods and as near-zero estimates for Bayesian DSEM.
eigen_tol	Numeric scalar. Estimated heterogeneity covariance matrices with minimum eigenvalues no greater than this value are classified as near singular.

Value

The output is saved as an external file in output_folder and is returned invisibly.

SumFitMetaVARLRT *Summary: MetaVAR Boundary Likelihood Ratio Test*

Description

Applies LRTMetaVAR() to every admissible MetaVAR replication of a task and summarizes the rejection rates at $\alpha = .05$.

Usage

```
SumFitMetaVARLRT(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARNormal *Summary: Normal Theory (FitMetaVAR)*

Description

Summary: Normal Theory (FitMetaVAR)

Usage

```
SumFitMetaVARNormal(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARRobust *Summary: Robust (FitMetaVAR)*

Description

Summary: Robust (FitMetaVAR)

Usage

```
SumFitMetaVARRobust(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMplus *Summary (FitMplus)*

Description

Summary (FitMplus)

Usage

```
SumFitMplus(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SumFitMplusDiagnostics`
Summary (FitMplus Diagnostics)

Description

Summary (FitMplus Diagnostics)

Usage

```
SumFitMplusDiagnostics(
  taskid,
  reps,
  output_folder,
  overwrite,
  integrity,
  ncores
)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`ncores`	Positive integer. Number of cores to use.

Details

This function is executed via the `Sum` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMplusPriors *Summary (FitMplus (User Defined Priors))*

Description

Summary (FitMplus (User Defined Priors))

Usage

```
SumFitMplusPriors(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMplusPriorsDiagnostics *Summary (FitMplus Diagnostics with User Defined Priors)*

Description

Summary (FitMplus Diagnostics with User Defined Priors)

Usage

```
SumFitMplusPriorsDiagnostics(
  taskid,
  reps,
  output_folder,
  overwrite,
  integrity,
  ncores
)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitNaive *Summary: Naive*

Description

Summary: Naive

Usage

```
SumFitNaive(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumStage1Calibration *Summary: Calibration of Stage 1 Sampling Variances*

Description

For each admissible MetaVAR replication of a task, computes the ratio of the between-person variance of the Stage 1 transition estimates to the mean Stage 1 sampling variance. Under zero transition heterogeneity, a ratio of one means that the Stage 1 sampling variances are calibrated; a ratio above one means that they are too small.

Usage

```
SumStage1Calibration(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the `Sum` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SumTolerance` *Summary: Tolerance Rates for Transition Heterogeneity*

Description

Reads the replication-level summaries of one task (MetaVAR with normal and robust intervals, the uncertainty-uncorrected method, and DSEM with default and user-specified priors) and computes, for each tolerance, the proportions of replications returned by `Tolerance()` for the between-person variances and covariances of the transition parameters.

Usage

```
SumTolerance(
  taskid,
  reps,
  folder,
  delta = c(0, 1e-04, 0.00025, 4e-04, 5e-04, 9e-04, 0.001, 0.0025, 0.005, 0.01)
)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`folder`	Character string. Folder with the summary files of the task.
`delta`	Numeric vector. Tolerances.

Value

A data frame with one row per method, interval type, parameter, and tolerance.

Author(s)

Ivan Jacob Agaloos Pesigan

TabMplusDiagSensitivity

Bayesian Diagnostic Failure-Rate Sensitivity Table

Description

Create a simulation-case-level table of Bayesian Mplus diagnostic failure rates. Unlike the overview figure, the function retains one row for every simulation task ID and does not average across simulation cases.

Usage

```
TabMplusDiagSensitivity(
  diagnostics,
  comparison = "difference",
  diagnostic = "Any diagnostic",
  unit = "replication",
  taskid = NULL,
  target = NULL,
  rhat_threshold = 1.01,
  ess_threshold = 400,
  mcse_threshold = 0.05,
  relative_mcse = TRUE,
  output = "data.frame",
  scale = "percent",
  digits = 0L
)
```

Arguments

diagnostics	An all-task diagnostics object containing a parameters data frame, or the parameter-level data frame itself.
comparison	Character string. "difference" returns paired user-prior minus default-prior failure-rate differences. "levels" returns the two prior-specific failure rates. "all" returns the default rate, user-prior rate, and paired difference.
diagnostic	One or more diagnostic panels. Available values are "Any diagnostic", "R-hat", "Bulk ESS", "Tail ESS", and "Relative MCSE" when relative_mcse = TRUE.
unit	Character string controlling the unit classified as failed. "replication" classifies a replication as failed when any parameter in the selected block fails. "parameter" averages failures over parameter-replication pairs.
taskid	Optional positive integer task IDs to include.
target	Optional parameter blocks to include. Available values are "Innovation", "FE", and "RE".
rhat_threshold	Numeric R-hat threshold.
ess_threshold	Numeric bulk and tail effective sample size threshold.

mcse_threshold	Numeric Monte Carlo standard error threshold.
relative_mcse	Logical. If TRUE, Monte Carlo standard errors are divided by the posterior standard deviation before classifying failures.
output	Character string. "data.frame" returns a wide data frame with simulation-design columns, "matrix" returns only the numeric case-by-summary matrix, and "long" returns one row per task, block, diagnostic, and prior comparison.
scale	Character string. "proportion" returns values on the 0-to-1 scale. "percent" multiplies prior-specific failure rates by 100 and expresses differences in percentage points.
digits	Nonnegative integer used to round the returned values.

Details

Failure rates are computed separately within each task ID. For comparison = "difference", replications are paired before calculating the user-prior minus default-prior difference. Negative differences favor the user-specified priors.

The default unit = "replication" answers the question: within a given simulation case, what percentage of replications had at least one failed parameter in the selected parameter block?

Value

A data frame or numeric matrix. For output = "matrix", simulation design information is stored in the "case_data" attribute.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
## Not run:
data(diagnostics, package = "manMetaVAR")

# Compact sensitivity matrix: one row per simulation case
TabMplusDiagSensitivity(
  diagnostics = diagnostics,
  comparison = "difference",
  output = "matrix"
)

# Default, user-prior, and difference columns
TabMplusDiagSensitivity(
  diagnostics = diagnostics,
  comparison = "all",
  output = "data.frame"
)

# Diagnostic-specific table
TabMplusDiagSensitivity(
```

```
  diagnostics = diagnostics,
  comparison = "difference",
  diagnostic = c("R-hat", "Bulk ESS", "Tail ESS", "Relative MCSE")
)

## End(Not run)
```

`Tolerance` *Tolerance Decisions for Between-Person Variances and Covariances*

Description

Classifies estimates and interval limits of between-person variances and covariances against a tolerance δ instead of exact zero. Variances are compared one-sided with δ; covariances two-sided with $[-\delta, \delta]$.

Usage

```
Tolerance(est, ll, ul, variance, delta)
```

Arguments

`est`	Numeric vector. Estimates.
`ll`	Numeric vector. Lower interval limits.
`ul`	Numeric vector. Upper interval limits.
`variance`	Logical vector. `TRUE` for variances, `FALSE` for covariances.
`delta`	Nonnegative numeric value. Tolerance.

Value

A data frame with logical columns `est_within` (estimate within the tolerance), `equivalent` (interval within the tolerance: practical homogeneity), and `exceed` (interval beyond the tolerance: heterogeneity beyond the tolerance).

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
Tolerance(
  est = c(0.0002, 0.01),
  ll = c(0, 0.004),
  ul = c(0.0008, 0.02),
  variance = c(TRUE, TRUE),
  delta = 0.0009
)
```

`tolerance_rates` *Tolerance Rates for Transition Heterogeneity*

Description

Proportions of replications in which the estimate or interval of a between-person variance or covariance of the transition parameters falls within or beyond a tolerance δ, computed with `SumTolerance()` from the replication-level simulation summaries.

Usage

```
data(tolerance_rates)
```

Format

A data frame with one row per design condition, method, interval type, parameter, and tolerance:

taskid Simulation task ID.

replications Number of replications.

replications_used Number of admissible replications.

method Estimation method, labeled as in results.

ci Confidence or credible interval method.

n Number of persons.

time Number of measurement occasions; `NA` denotes the unbalanced condition.

heterogeneity Transition-parameter heterogeneity multiplier: 0, 1, or 2.

parnames Parameter name.

parameter Population value.

type `"variance"` or `"covariance"`.

delta Tolerance.

est_within Proportion of estimates within the tolerance.

equivalent Proportion of intervals within the tolerance (practical homogeneity).

exceed Proportion of intervals beyond the tolerance (heterogeneity beyond the tolerance).

Author(s)

Ivan Jacob Agaloos Pesigan

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